

WEEK – 5 DBMS TASK

PAYMENT TRANSACTION MANAGEMENT SYSTEM

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TASK DESCRIPTION

1. Create Customer and Payment tables.
2. Store customer and payment transaction details.
3. Record successful, pending, and failed transactions.
4. Analyze payment modes and transaction amounts.
5. Generate customer-wise and daily payment reports.

SQL CODE

```
-- =====
-- WEEK 5 DBMS TASK
-- PAYMENT TRANSACTION MANAGEMENT SYSTEM
-- =====

CREATE DATABASE PaymentManagementDB;

USE PaymentManagementDB;

-- =====
-- 1. CREATE CUSTOMER TABLE
-- =====

CREATE TABLE Customer (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    phone VARCHAR(15)
);
```

```

-- =====

-- 2. CREATE PAYMENT TABLE

-- =====

CREATE TABLE Payment (

    payment_id INT PRIMARY KEY,

    customer_id INT,

    order_id INT,

    payment_mode VARCHAR(30) NOT NULL,

    payment_date DATE NOT NULL,

    amount DECIMAL(10,2) NOT NULL,

    payment_status VARCHAR(20) NOT NULL,

    FOREIGN KEY (customer_id)

        REFERENCES Customer(customer_id)

);

-- =====

-- 3. INSERT CUSTOMERS

-- =====

INSERT INTO Customer VALUES

(1, 'Arun Kumar', 'arun@gmail.com', '9000001001'),

(2, 'Meena Devi', 'meena@gmail.com', '9000001002'),

(3, 'Karthik Raj', 'karthik@gmail.com', '9000001003'),

(4, 'Divya Sri', 'divya@gmail.com', '9000001004'),

(5, 'Naveen Kumar', 'naveen@gmail.com', '9000001005');

-- =====

-- 4. INSERT PAYMENT TRANSACTIONS

-- =====

INSERT INTO Payment VALUES

(601, 1, 2001, 'UPI', '2026-08-20', 12500.00, 'Successful'),

(602, 2, 2002, 'Credit Card', '2026-08-21', 18500.00, 'Successful'),

```

```
(603, 3, 2003, 'Debit Card', '2026-08-22', 4200.00, 'Pending'),
(604, 4, 2004, 'Net Banking', '2026-08-23', 9600.00, 'Failed'),
(605, 5, 2005, 'UPI', '2026-08-24', 7200.00, 'Successful'),
(606, 1, 2006, 'Cash', '2026-08-25', 3000.00, 'Successful'),
(607, 3, 2007, 'UPI', '2026-08-26', 15500.00, 'Failed'),
(608, 2, 2008, 'Credit Card', '2026-08-27', 8300.00, 'Successful');
```

```
-- =====
```

```
-- 5. DISPLAY ALL CUSTOMERS
```

```
-- =====
```

```
SELECT * FROM Customer;
```

```
-- =====
```

```
-- 6. DISPLAY ALL PAYMENTS
```

```
-- =====
```

```
SELECT * FROM Payment;
```

```
-- =====
```

```
-- 7. DISPLAY SUCCESSFUL PAYMENTS
```

```
-- =====
```

```
SELECT *
```

```
FROM Payment
```

```
WHERE payment_status = 'Successful';
```

```
-- =====
```

```
-- 8. DISPLAY FAILED PAYMENTS
```

```
-- =====
```

```
SELECT *
```

```
FROM Payment
```

```
WHERE payment_status = 'Failed';
```

```
-- =====
```

```
-- 9. DISPLAY PENDING PAYMENTS
```

```
-- =====
```

```
SELECT *  
  
FROM Payment  
  
WHERE payment_status = 'Pending';
```

```
-- =====
```

```
-- 10. UPDATE A PENDING PAYMENT
```

```
-- =====
```

```
UPDATE Payment  
  
SET payment_status = 'Successful'  
  
WHERE payment_id = 603;
```

```
-- =====
```

```
-- 11. PAYMENT DETAILS WITH CUSTOMER NAME
```

```
-- =====
```

```
SELECT  
  
    p.payment_id,  
  
    c.customer_name,  
  
    p.order_id,  
  
    p.payment_mode,  
  
    p.payment_date,  
  
    p.amount,  
  
    p.payment_status  
  
FROM Payment p  
  
JOIN Customer c  
  
    ON p.customer_id = c.customer_id  
  
ORDER BY p.payment_date;
```

```
-- =====  
  
-- 12. COUNT TRANSACTIONS BY PAYMENT MODE  
  
-- =====  
  
SELECT  
  
    payment_mode,  
  
    COUNT(*) AS total_transactions  
  
FROM Payment  
  
GROUP BY payment_mode  
  
ORDER BY total_transactions DESC;
```

```
-- =====  
  
-- 13. TOTAL SUCCESSFUL AMOUNT BY PAYMENT MODE  
  
-- =====  
  
SELECT  
  
    payment_mode,  
  
    SUM(amount) AS total_amount  
  
FROM Payment  
  
WHERE payment_status = 'Successful'  
  
GROUP BY payment_mode  
  
ORDER BY total_amount DESC;
```

```
-- =====  
  
-- 14. TRANSACTION COUNT BY STATUS  
  
-- =====  
  
SELECT  
  
    payment_status,  
  
    COUNT(*) AS total_transactions  
  
FROM Payment  
  
GROUP BY payment_status;
```

```
-- =====  
-- 15. TOTAL SUCCESSFUL PAYMENT AMOUNT
```

```
-- =====  
  
SELECT  
  
    SUM(amount) AS total_successful_amount  
  
FROM Payment  
  
WHERE payment_status = 'Successful';
```

```
-- =====  
-- 16. TOTAL FAILED PAYMENT AMOUNT
```

```
-- =====  
  
SELECT  
  
    SUM(amount) AS total_failed_amount  
  
FROM Payment  
  
WHERE payment_status = 'Failed';
```

```
-- =====  
-- 17. HIGH-VALUE PAYMENTS ABOVE 10000
```

```
-- =====  
  
SELECT  
  
    payment_id,  
  
    customer_id,  
  
    order_id,  
  
    payment_mode,  
  
    amount,  
  
    payment_status  
  
FROM Payment  
  
WHERE amount > 10000  
  
ORDER BY amount DESC;
```

```
-- =====
```

```
-- 18. CUSTOMER-WISE PAYMENT REPORT
```

```
-- =====
```

```
SELECT
```

```
    c.customer_id,
```

```
    c.customer_name,
```

```
    COUNT(p.payment_id) AS total_payments,
```

```
    SUM(
```

```
        CASE
```

```
            WHEN p.payment_status = 'Successful'
```

```
            THEN p.amount
```

```
            ELSE 0
```

```
        END
```

```
    ) AS successful_amount
```

```
FROM Customer c
```

```
LEFT JOIN Payment p
```

```
    ON c.customer_id = p.customer_id
```

```
GROUP BY c.customer_id, c.customer_name
```

```
ORDER BY successful_amount DESC;
```

```
-- =====
```

```
-- 19. DAILY PAYMENT REPORT
```

```
-- =====
```

```
SELECT
```

```
    payment_date,
```

```
    COUNT(*) AS total_transactions,
```

```
    SUM(amount) AS total_amount
```

```
FROM Payment
```

```
GROUP BY payment_date
```

```
ORDER BY payment_date;
```

```
-- =====  
-- 20. COMPLETE PAYMENT TRANSACTION REPORT  
-- =====  
  
SELECT  
    p.payment_id,  
    c.customer_name,  
    p.order_id,  
    p.payment_mode,  
    p.payment_date,  
    p.amount,  
    p.payment_status  
FROM Payment p  
JOIN Customer c  
    ON p.customer_id = c.customer_id  
ORDER BY p.payment_id;
```